

OBJECTIVES

- Understand the design and operation of a common-source (CS) amplifier.

EQUIPMENT

- 2 each: 2N7000 NMOS FET
- Multiple capacitors
- Various resistors
- Breadboard and various wires
- Digilent Analog Discovery (DAD) Board

INTRODUCTION

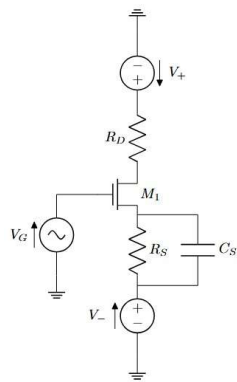


Figure 1 – MOSFET amplifier

- **Drain Resistance**

In this lab we will manipulate the drain resistor ($R_D=220\ \Omega$) in circuits like the one above to perform certain functions. For this lab, this includes measuring drain current (I_D) and serving as the load resistor in a common-source amplifier.

- **Testing/Tools**

We will test the circuits two ways:

- With the waveform generator as a DC voltage source
- With a sine wave superimposed on a DC bias voltage.

We will use the scope in two ways as well:

- As a DC voltmeter
- In XY mode to plot V_{out} vs. V_{in}

- **Structure**

We will first characterize the current-voltage behavior of your MOSFET and then choose a good operating point for the circuit as an amplifier. Then we will characterize the circuit as an amplifier. Finally, we will add a circuit using a negative power supply to set a more stable bias current and characterize the improved circuit. All the circuits will be built using variations on same setup in figure

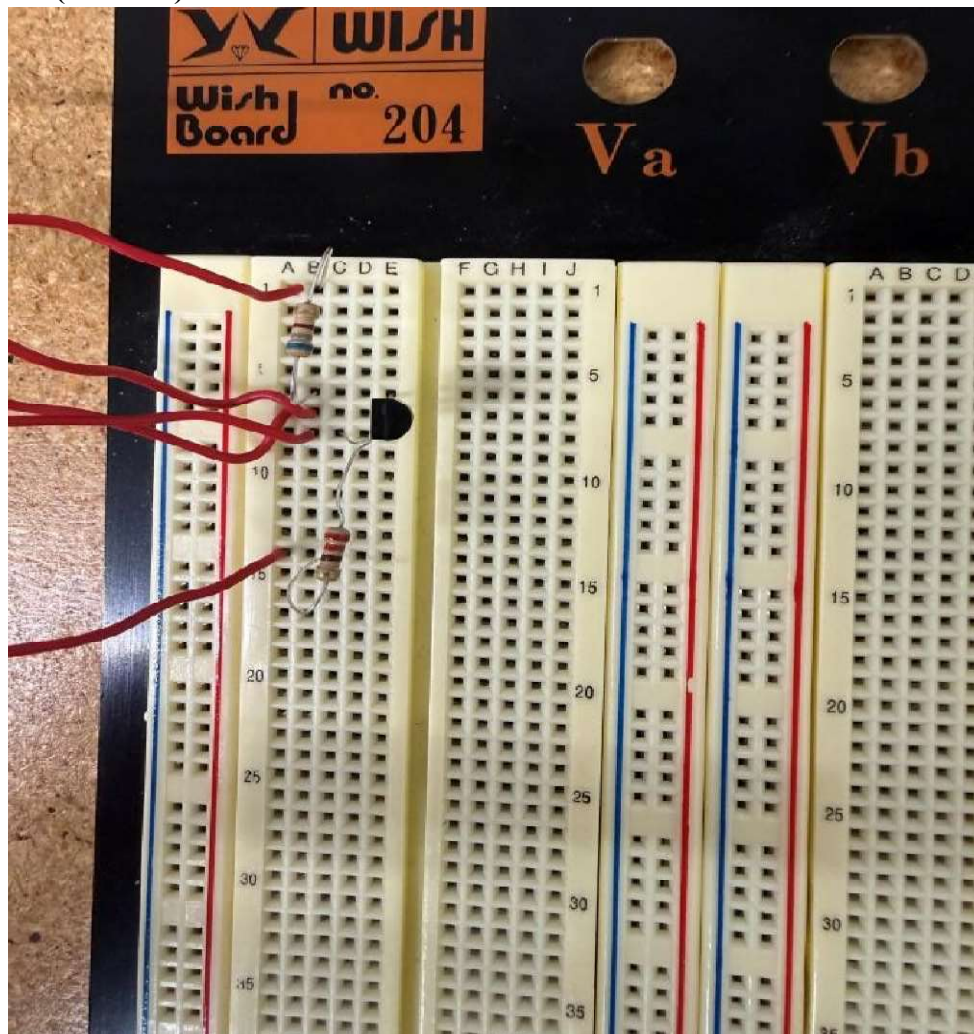
Lab 4: MOSFET Amplifiers – In-Lab

In-Lab

Part 1: DC-Degenerated Common-Source Amplifier (20 Points)

The amplifier requires a carefully selected DC bias voltage to be added to the input signal. A different FET would likely require a different value. To make the current less dependent on MOSFET parameter variations, we can use a circuit like the one shown below using the negative power supply and a “degeneration” resistor $R_{S1} = V_{SS} / I_{DQ}$ (previous current)

- I. Build the following circuit in Figure 2 on your breadboard using a 2N7000 Transistor and a $R_{S1} = 680 \Omega$, $R_D = 220 \Omega$. (4 Points)



Lab 4: MOSFET Amplifiers – In-Lab

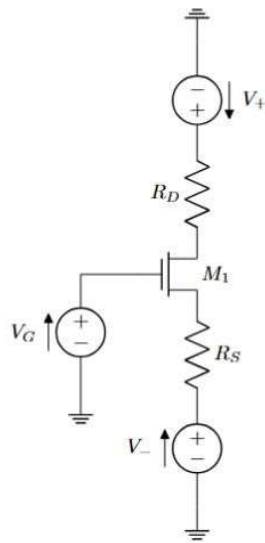
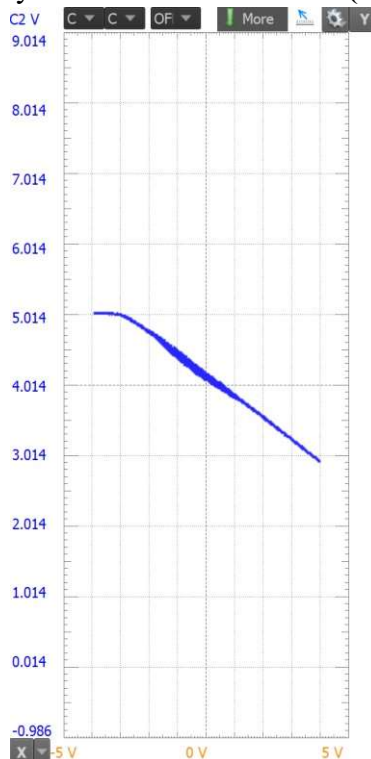


Figure 2 – Common-source amplifier with degeneration.

- II. Measure the effect of R_{S1} on the voltage transfer curve. Set up WaveGen1 to supply a 10 kHz sinusoid with peak to peak voltage from 1 V to 4 V. Set up the scope in XY mode with Channel 1 (V_G) on the x-axis and Channel 2 (V_D) on the y-axis. Adjust the input offset and amplitude and the scope sensitivities to get a nice VTC display and insert a screenshot. **(2 Points)**



Lab 4: MOSFET Amplifiers – In-Lab

- III. Using the same setup, swap your 2N7000 for a different one and plot the resulting VTC. Compare the sensitivity to swapping the FET. How does including R_S affect the sensitivity to the FET parameters compared to when it wasn't included in the prelab? **(2 Points)**

It makes the switch of the FETS not different as it did in the pre-lab.



Lab 4: MOSFET Amplifiers – In-Lab

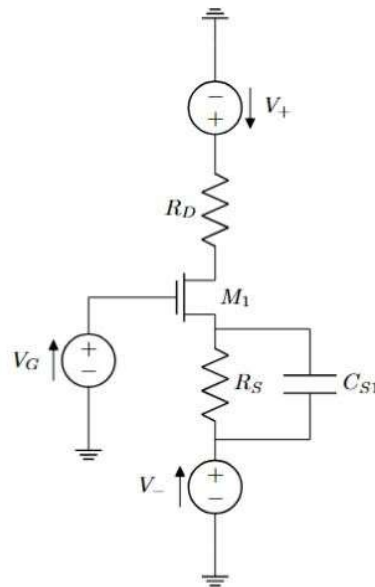
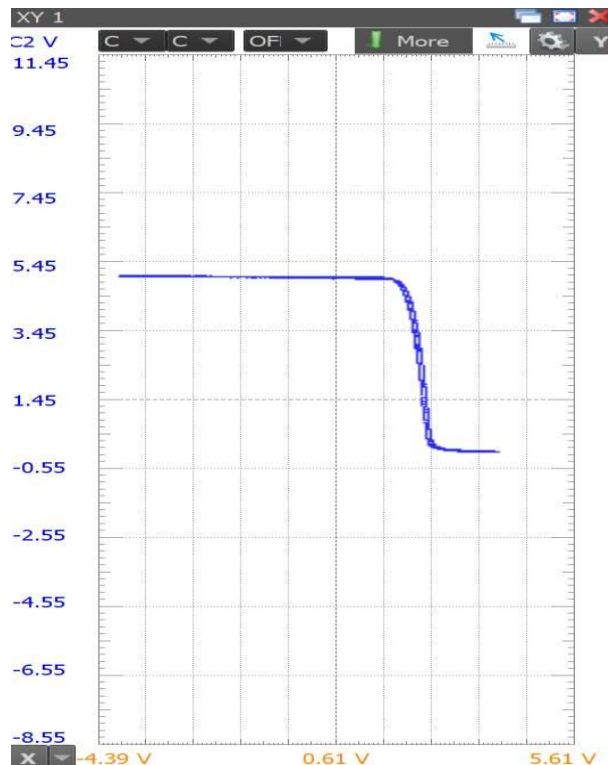


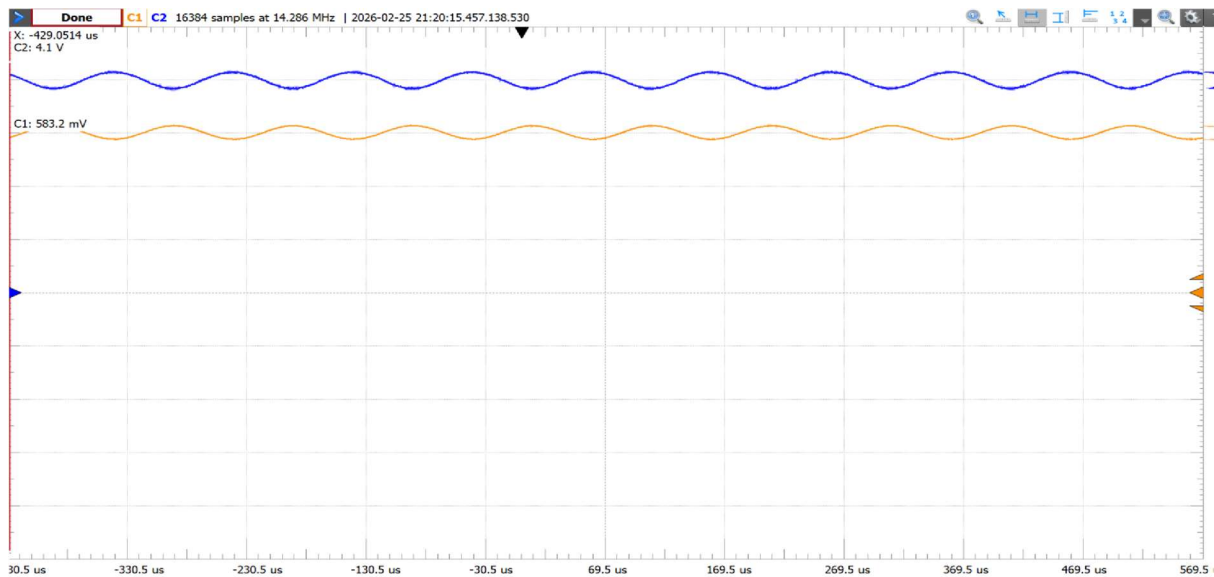
Figure 3 – Common-source amplifier with degeneration (with Bypass capacitor).

- IV. Including R_S reduces (*degenerates*) the voltage gain. To reobtain the voltage gain, add a large-valued bypass capacitor C_{S1} in parallel with R_{S1} . We'll make C_{S1} 's value and the input frequency large enough that C_{S1} won't have time to charge or discharge much during half cycles of the input. Add $C_{S1} = \sim 22 \mu\text{F}$. Electrolytic capacitors are polarized; connect the + terminal of your capacitor to the higher voltage, not to V_{SS} . **Obtain your VTC. (4 Points)**



- V. With WaveGen1, apply a 50 mV 10 kHz sine wave with DC offset = V_{GQ} and for V_G . * If the output signal doesn't look sinusoidal, reduce the input signal amplitude. **Measure the amplitudes of the input and output sinusoids v_o and v_i . Save these waveforms for your report. Compute the voltage gain $A_v = v_o / v_i$. (4 Points)**

$V_i = 55.147\text{mV}_{pp}$, $V_o = 342.69\text{mV}_{pp}$, $A_v = 342.69/55.147 = 6.21$



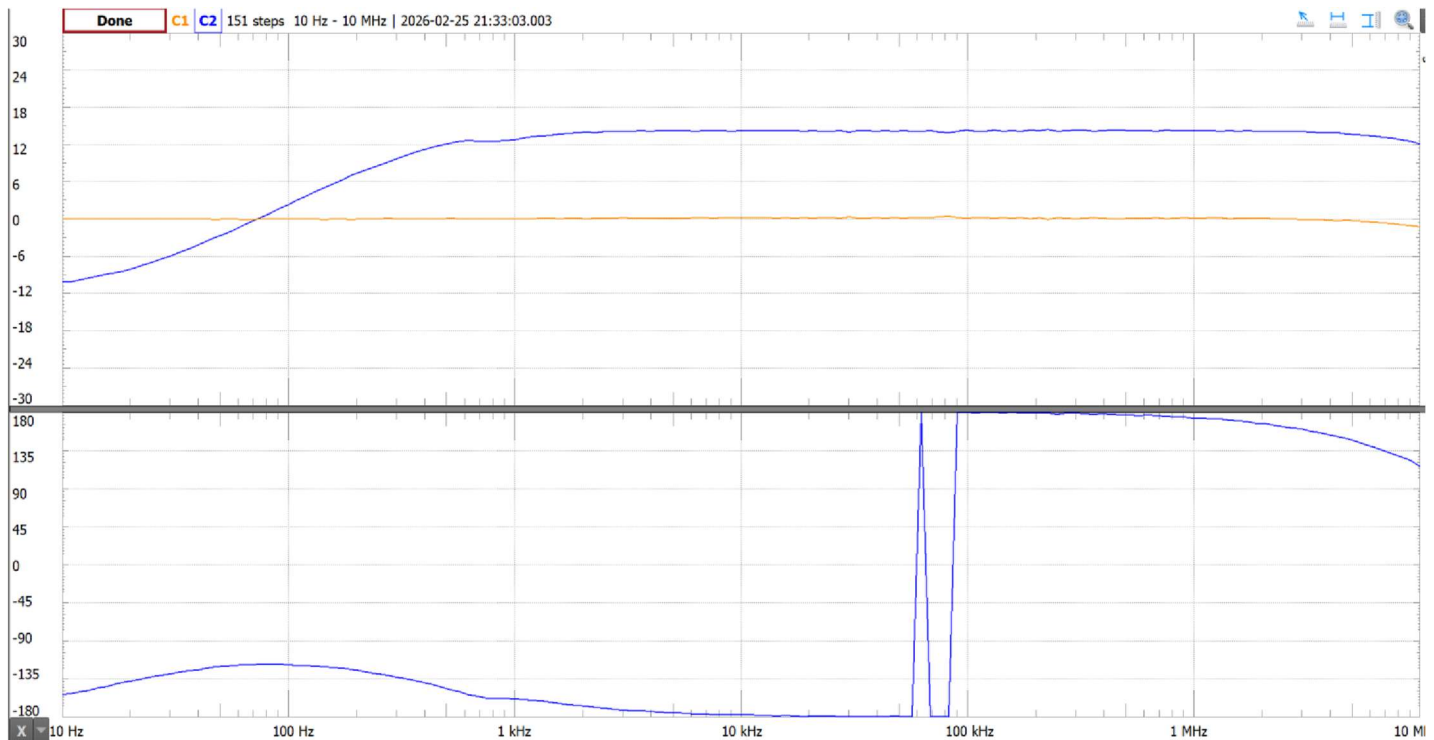
Lab 4: MOSFET Amplifiers – In-Lab

- VI. Use the network analyzer to obtain the frequency response of the circuit. Identify the frequency (f_z) at which the gain enhancement due to the capacitor begins and the frequency f_L where the gain reaches its maximum (mid-band) value. Also note the upper cutoff (-3dB) frequency f_H . (3 Points)

F_z = Around 25Hz

F_L = Around 500Hz

F_H = Around 7-8MHz



- VII. What would happen to frequencies f_z and f_L if you used a higher-value capacitor for C_{S1} . (1 Point)

They will both go to lower frequencies as we increase C_{S1}

Part 2: AC-Coupled Common-Source Amplifier (3 Points) (Optional)

In this part, the amplifier is modified to use AC coupling at both the input and output while establishing the MOSFET's DC gate bias with a resistor voltage divider. This configuration allows the amplifier to process small AC signals without disturbing the chosen operating point. This improves signal integrity and makes the circuit more practical for real-world applications.

- I. Build the following circuit in Figure 3 on your breadboard using a 2N7000 Transistor, $R_{S1} = 680\ \Omega$, $R_D = 220\ \Omega$, $C_{D1} = 1\ \mu\text{F}$, $C_{G1} = 1\ \mu\text{F}$, and $C_{S1} = 22\ \mu\text{F}$. Choose values for R_{B1} and R_{B2} to set $V_G = V_{GQ}$, while keeping their parallel combination $> 100\ \text{k}\Omega$. **(1 Point)**

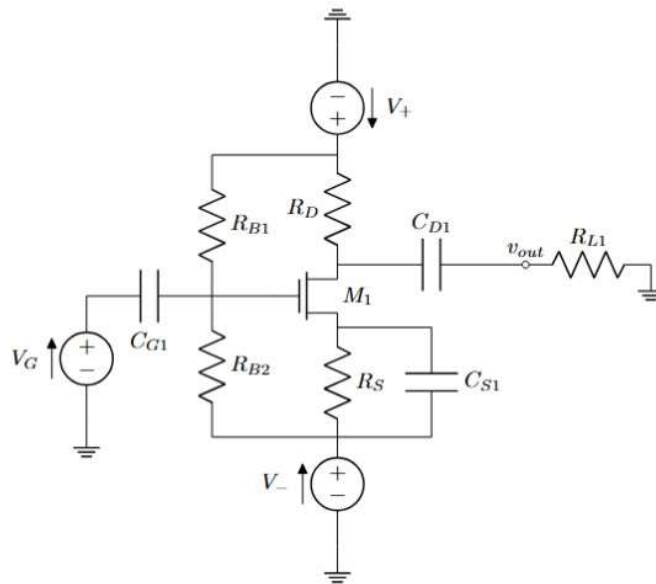


Figure 4 – AC-coupled common-source amplifier.

- II. Put WaveGen1 in DC mode and slightly shift V_G while observing V_G and V_D . If it's working right, the circuit should return to your operating point after every change. Apply a 50 mV 10 kHz sine wave for V_G . As before, if the output signal doesn't look sinusoidal, reduce the input signal amplitude. **(1 Point)**
- III. Measure the amplitudes of the input and output sinusoids. Take note of these waveforms for your report. Compute the mid-band voltage gain. **(1 Point)**